On Graphical Minimal Surfaces and the Homotheticity of Singular Behavior in Mean Curvature Flow

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Context

Question

If a wire loop is suspended in space, what is the most "efficient" surface that spans it?

Physicist Joseph Plateau first explored this by dipping wire loops into soapy water. This motivated "Plateau's Problem," a question about the conditions for the existence of a minimal surface for a given boundary condition. Before exploring this, however, we must develop the necessary machinery.

First and Second Fundamental Forms

We denote the first fundamental form as the symmetric 2-tensor

$$g = \begin{bmatrix} \left\langle \frac{\partial F}{\partial u_1}, \frac{\partial F}{\partial u_1} \right\rangle & \left\langle \frac{\partial F}{\partial u_1}, \frac{\partial F}{\partial u_2} \right\rangle \\ \\ \left\langle \frac{\partial F}{\partial u_2}, \frac{\partial F}{\partial u_1} \right\rangle & \left\langle \frac{\partial F}{\partial u_2}, \frac{\partial F}{\partial u_2} \right\rangle \end{bmatrix},$$

and the second fundamental form as

$$h = \begin{bmatrix} \left\langle \frac{\partial^2 F}{\partial u_1 \partial u_1}, \nu \right\rangle & \left\langle \frac{\partial^2 F}{\partial u_1 \partial u_2}, \nu \right\rangle \\ \left\langle \frac{\partial^2 F}{\partial u_2 \partial u_1}, \nu \right\rangle & \left\langle \frac{\partial^2 F}{\partial u_2 \partial u_2}, \nu \right\rangle \end{bmatrix}.$$

Standard theory shows that the normal curvature in a direction $\hat{e}$ at a point p is simply $h(\hat{e}, \hat{e})$.

Principal Curvatures

To find the maximum and minimum directions of normal curvature, we use Lagrange multipliers:

$$\nabla h(\hat{e}, \hat{e}) = \lambda \nabla g(\hat{e}, \hat{e}),$$

$$g(\hat{e}, \hat{e}) = 1.$$

Solving yields,

$$h \begin{bmatrix} x^1 \\ x^2 \end{bmatrix} = \lambda g \begin{bmatrix} x^1 \\ x^2 \end{bmatrix}.$$

Therefore, the eigenvalues of the matrix $g^{-1}h$ are the principal (maximum and minimum) curvatures. We denote their average

$$H = \frac{\lambda_1 + \lambda_2}{2},$$

to be mean curvature.

Defining a Minimal Surface

Fix $\mathcal{M} \subseteq \mathbb{R}^3$ as a regular surface that spans Γ. Let $F(u_1, u_2)$ be a local parameterization of $\mathcal{M}$. Now consider a family of surfaces parametrized by

$$\tilde{F}(u_1, u_2, s) = F(u_1, u_2) + s\phi(u_1, u_2)\nu(u_1, u_2).$$

If $\mathcal{M} = \mathcal{M}(0)$ is minimal,

$$\frac{d}{ds}[\mathcal{M}(s)]\bigg|_{s=0} = 0.$$

It can be shown as a lemma that if $g(s)$ is a family of symmetric matrices with $\det g(s) > 0$. Then,

$$\frac{d}{ds} \det g(s) = \det g(s) \ \mathrm{tr}\left(g(s)^{-1} g(s)\right).$$

The area of a surface can be calculated by

$$[\mathcal{M}(s)] = \int \sqrt{\det g(s)} du_1 du_2.$$

So,

$$\frac{d}{ds}[\mathcal{M}(s)]\Big|_{s=0} = \int_{\mathcal{M}} \left(\frac{\partial}{\partial s} \sqrt{\det g(s)} \Big|_{s=0} \right) du_1 du_2.$$

Simplifying by applying the lemma, we obtain

Theorem (First Variation of Area)

$$\frac{d}{ds}[\mathcal{M}(s)]\Big|_{s=0} = -2 \int_{\mathcal{M}} \phi H \sqrt{\det g} du_1 du_2.$$

As ϕ is not uniformly zero and $\sqrt{\det g}$ is strictly positive, the variation is only uniformly zero if $\forall p \in \mathcal{M}, H(p) = 0$.

Graphical Minimal Surface PDE

In contrast to general boundary conditions, where existence cannot be affirmed, a Leray-Schauder fixed point theorem argument guarantees existence for $\Gamma \subseteq \mathbb{R}^3$, with $\mathcal{M} \subseteq \mathbb{R}^3$ as a graph of $f : \mathcal{U} \subseteq \mathbb{R}^2 \to \mathbb{R}$ parameterized by $F(x, y, f(x, y))$ such that $\partial\mathcal{M} = \Gamma$.

Such a setup yields the minimality PDE

$$\frac{\partial}{\partial x}\left(\frac{f_x}{\sqrt{1+|\nabla f|^2}}\right) + \frac{\partial}{\partial y}\left(\frac{f_y}{\sqrt{1+|\nabla f|^2}}\right) = 0.$$

Boundary conditions must still be considered separately.

Uniqueness of Graphical Minimal Surfaces

For graphically defined minimal surfaces, we have a guaranteed uniqueness property. To prove such, we first apply the following:

Theorem (Maximum Principle of Elliptic Equations)

Let $\Omega \subseteq \mathbb{R}^2$, and let $U(x_1, x_2) : \Omega \to \mathbb{R}$. Suppose

$$L(U)(x_1, x_2) = \sum_{i,j=1}^{2} a^{ij}(x_1, x_2) \frac{\partial^2 U}{\partial x_i \partial x_j} + \sum_{i=1}^{2} b^i(x_1, x_2) \frac{\partial U}{\partial x_i} = 0.$$

Also suppose the matrix

$$A = \begin{bmatrix} a^{11}(x, y) & a^{12}(x, y) \\ a^{21}(x, y) & a^{22}(x, y) \end{bmatrix}$$

is symmetric, positive definite. Then,

$$\max_{\Omega} U = \max_{\partial\Omega} U.$$

Proof of Maximum Principle

Let $\epsilon > 0$ be an arbitrary positive real number, and let

$$U_\epsilon(x_1, x_2) = U(x_1, x_2) + \epsilon e^{\delta x_1}, \delta > 0 \text{ to be chosen.}$$

Suppose for contradiction we can find a maximum $x_0 \in \Omega \setminus \partial\Omega$. By the first and second derivative tests, we then have,

$$\nabla U_\epsilon(x_0) = 0,$$

and that the matrix

$$D^2 U_\epsilon(x_0) = \begin{bmatrix} \frac{\partial^2 U_\epsilon}{\partial x_1 \partial x_1}(x_0) & \frac{\partial^2 U_\epsilon}{\partial x_1 \partial x_2}(x_0) \\ \\ \frac{\partial^2 U_\epsilon}{\partial x_2 \partial x_1}(x_0) & \frac{\partial^2 U_\epsilon}{\partial x_2 \partial x_2}(x_0) \end{bmatrix}$$

is negative semi-definite. Then, by inspection we see

$$\sum_{i,j=1}^{2} a^{ij}(x_0) \frac{\partial^2 U_\epsilon}{\partial x_i \partial x_j}(x_0) = \operatorname{tr}(A \cdot D^2 U_\epsilon(x_0)).$$

Proof of Maximum Principle Pt. 2

As we are multiplying a positive matrix by a non-positive matrix, it is clear that

$$\sum_{i,j=1}^{2} a^{ij}(x_0)\frac{\partial^2 U_\epsilon}{\partial x_i \partial x_j}(x_0) \leq 0.$$

As $L(U_\epsilon)(x_0)$ is the sum of a non-positive part and a zero part, it is less than or equal to zero. On the other hand,

$$\frac{\partial}{\partial x_1} e^{\delta x_1} = \delta e^{\delta x_1}, \frac{\partial}{\partial x_2} e^{\delta x_1} = 0, \frac{\partial^2}{\partial x_1^2} e^{\delta x_1} = \delta^2 e^{\delta x_1}, \frac{\partial^2}{\partial x_1 \partial x_2} e^{\delta x_1} = 0, \frac{\partial^2}{\partial x_2^2} e^{\delta x_1} = 0.$$

Thus,

$$L(e^{\delta x_1}) = \delta e^{\delta x_1}\left[a^{11}\delta + b^1\right] \geq \delta e^{\delta x_1}\left[a^{11}\delta - \max_{\Omega}|b^1|\right].$$

Proof of Maximum Principle Pt. 3

Since $a^{11}(x_1, x_2) > 0$ in Ω, we choose $\delta > 0$ sufficiently large so that

$$(\min_{\Omega} a^{11})\delta - \max_{\Omega} |b^1| > 0.$$

This implies, in Ω,

$$L(e^{\delta x_1}) > 0.$$

Thus, in Ω, we have

$$L(U_\epsilon) = L(U) + \epsilon L(e^{\delta x_1}) > 0,$$

yielding the absurdity,

$$0 < L(U_\epsilon)(x_0) \leq 0,$$

completing our contradiction.

Uniqueness of Graphical Minimal Surfaces Pt. 2

Then, we apply the following:

Lemma

If U_1 and U_2 are solutions to the minimal graph equation over Ω, then the function $W = U_1 - U_2$ satisfies an equation of the form

$$div(A \cdot \nabla W) = 0,$$

where A is a positive definite matrix.

The first part can be proved by defining a mapping $\chi : \mathbb{R}^2 \to \mathbb{R}^2$ as $\chi(\vec{x}) = \frac{\vec{x}}{\sqrt{1+|\vec{x}|^2}}$. and applying it to ∇U_1 and ∇U_2. The second part is more challenging and involves writing A as an integral involving the Jacobian of χ and then proving it's positive definite using Cauchy-Schwarz. Finally, we let U_1, U_2 be two solutions of the minimal graphical surface equation over Ω. The lemma above guarantees that they are identical within the region Ω.

Interesting Corollaries

As a result of uniqueness, we get two interesting corollaries.

Corollary

If a boundary curve $\Gamma \in \Omega$ is contained in a plane π, then all minimal graphs $\mathcal{M} \in \Omega$ are likewise contained in π.

The proof is simply by showing that the general equation for a plane satisfies the minimal surface PDE with a planar boundary condition. Uniqueness guarantees it is the only possible form.

Corollary

Let Ω be a unit disk and $\mathcal{M} \in \Omega$ a minimal surface with $\partial\mathcal{M} = \Gamma$. If Γ is rotationally symmetric, then $\mathcal{M}$ is rotationally symmetric.

The proof involves letting

$$\varphi(y_1, y_2) = \mathcal{M}\left(\cos\alpha y_1 - \sin\alpha y_2, \sin\alpha y_1 + \cos\alpha y_2\right) = \mathcal{M}(x_1, x_2).$$

It then suffices to show that $\varphi(y_1, y_2) = \mathcal{M}(y_1, y_2)$, through involved calculation.

Mean Curvature Flow

We have studied static surfaces, but it is also interesting to look at evolving surfaces. Consider the first variation of area, this time in the form:

$$-2\int_{\mathcal{M}} \left\langle \frac{\partial F_s}{\partial s}, \nu \right\rangle H\sqrt{\det g}\, du_1\, du_2.$$

Considering a variation,

$$\left\langle \frac{\partial F_s}{\partial s}, \nu \right\rangle = H(F_s),$$

$$\frac{d}{ds}[\mathcal{M}(s)]$$

is monotonically non-increasing. By such, it is useful to define the following:

Definition (Mean Curvature Flow)

$$\begin{cases} \left(\frac{\partial F_t}{\partial t}\right)^{\perp} = H\nu \\ F_0 = F \end{cases}$$

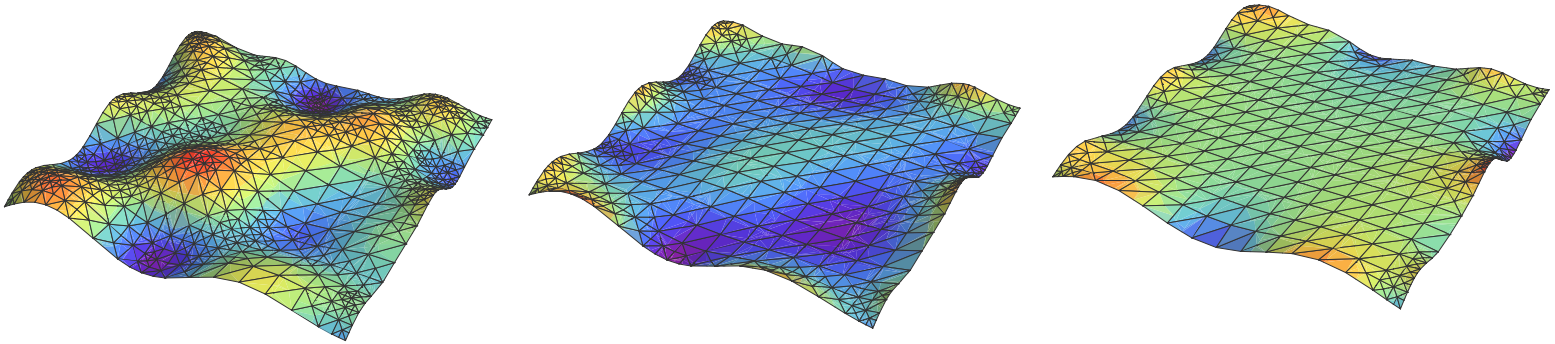
Approximations With Mean Curvature Flow

Definition (Graphical Mean Curvature Flow)

$$\begin{cases} \frac{\partial \mathcal{M}}{\partial t} = \sqrt{1+|\nabla \mathcal{M}|^2}\ \mathrm{div}\left(\frac{\nabla \mathcal{M}}{\sqrt{1+|\nabla \mathcal{M}|^2}}\right) & \text{in } \Omega \times (0,\infty) \\ \mathcal{M}(x,y,t) = \Gamma(x,y) & \text{on } \partial\Omega \times (0,\infty) \\ \mathcal{M}(x,y,0) = \rho(x,y) & \text{in } \bar{\Omega} \times \{0\} \end{cases} \text{ for}$$

some function $\rho(x,y) : \Omega \longrightarrow \mathbb{R}$ such that $\rho = \Gamma$ on $\partial\Omega$.

As a unique graphical minimal surface necessarily exists for a given graphical boundary condition, the flow will reduce to approximate the minimal surface. This can be especially useful for complicated boundary conditions with no analytic solution.



Self-Shrinkers and Type I Singularities

Often, curvature flows can exhibit singularities categorized by the rate at which curvature flows up. For type I singularities,

$$\exists C : \sup_{\mathcal{M}_t} |h|^2(x,t) \leq \frac{C}{T-t},$$

for $t \in (0, T)$.

Theorem (Self-shrinkers as a Model for Type I Singularities)

Let $\{\mathcal{M}_t\}_{t<T} \subset \mathbb{R}^{n+1}$ be a smooth mean curvature flow. Suppose (x_0, T) is a type I singularity. Then, for all sequences $\lambda_k \to \infty$, the rescaled flows

$$\mathcal{M}_s^{(k)} := \lambda_k \left(\mathcal{M}_{T+\lambda_k^{-2}s} - x_0 \right), s < 0$$

sub-sequentially converge smoothly to a limit flow $\tilde{\mathcal{M}}_s$, which satisfies the self-shrinker equation $\vec{H} = \frac{1}{2} \langle x, \nu \rangle$.

Proof of Main Theorem

First we need:

Theorem (Huisken's Monotonicity Formula)

$$\frac{\partial}{\partial t}\int_{\mathcal{M}_t}\Phi = -\int_{\mathcal{M}_t}\left|\vec{H} - \frac{\nabla^{\perp}\Phi}{\Phi}\right|^2 \Phi dVol,$$

where Φ is the backwards heat kernel.

Recall the backwards heat kernel,

$$\Phi(x,t) = \frac{1}{(-4\pi t)^{\frac{n}{2}}} e^{\frac{|x|^2}{4t}},$$

Then we look to prove:

Lemma

Let $\{\mathcal{M}_t\}_{t\in[0,T)}$ be a mean curvature flow with $|h| \leq C$ for $t \in [0,T)$. Then, $\exists C_1(C,T)$ such that $|\nabla h| \leq C_1$ on $[\frac{T}{2}, T)$.

Proof of Main Theorem Pt. 2

We look to prove this lemma through a Shi-type estimate. To start, we consider $\varsigma(t,x) = \alpha|h|^2 + t|\nabla h|^2$, for some $\alpha > 0$. Now, we compute

$$\left(\frac{\partial}{\partial t} - \Delta_{\mathbb{R}^{n+1}}\right)\varsigma = \alpha\left(\frac{\partial}{\partial t} - \Delta_{\mathbb{R}^{n+1}}\right)|h|^2 + |\nabla h|^2 + t\left(\frac{\partial}{\partial t} + \Delta_{\mathbb{R}^{n+1}}\right)|\nabla h|^2$$

$$= -2\alpha|\nabla h|^2 + 2\alpha|h|^4 + |\nabla h|^2 + t\left(-2|\nabla^2 h|^2 + \epsilon\right),$$

where ϵ is an error term comprised of low order terms.
Now, by the assumption of a bounded curvature estimate and work to show that the error is bounded, we have the inequality

$$\left(\frac{\partial}{\partial t} - \Delta_{\mathbb{R}^{n+1}}\right)\varsigma \leq 2\alpha C^4 + \left(1 - 2\alpha + tDC^2\right)|\nabla h|^2.$$

Now, we pick α such that for our largest case, $t = T$,

$$\frac{1 + TDC^2}{2} < \alpha.$$

Proof of Main Theorem Pt. 3

Thus,

$$\left(\frac{\partial}{\partial t} - \Delta_{\mathbb{R}^{n+1}}\right)\varsigma \leq 2\alpha C^4.$$

Now, by the parabolic maximum principle,

$$\varsigma(t,x) \leq 2\alpha C^4 T + \varsigma(0,x)$$

$$\leq 2\alpha C^4 T + \alpha C^2.$$

Now, recalling our definition for $\varsigma(t,x)$, we have

$$t|\nabla h|^2 \leq 2\alpha C^4 T + \alpha C^2, \forall t \in [0, T).$$

Thus, we have an a priori bound

$$|\nabla h| \leq C_1(T, C), \forall t \in [\frac{T}{2}, T).$$

By recursion, this bounds every derivative.

Proof of Main Theorem Pt. 3.5

Finally we have built the machinery to kill the result! After a parabolic rescaling, we have

$$T - t = -\lambda_k^{-2} s.$$

So,

$$|h_{\mathcal{M}_s^{(k)}}| = \frac{1}{\lambda_k}|h_{\mathcal{M}_t}| \leq \frac{1}{\lambda_k}\frac{C}{\sqrt{T-t}} = \frac{C}{-s}, \forall k,$$

for a type I singularity. As curvature and its derivatives are bounded, standard compactness theorems say that

$$\exists \{k_n\} \to \infty : \mathcal{M}_s^{(k_n)} \xrightarrow{C^\infty_{\text{loc}}} \tilde{\mathcal{M}}_s.$$

Huisken's Formula gives

$$\frac{d}{dt}\theta_{x_0,T}(t) = -\int_{\mathcal{M}_t} \left|\vec{H} + \frac{(x-x_0)^\perp}{2(T-t)}\right|^2 \Phi(x,t)\text{dVol}.$$

Consider $t_1 < t_2$ and integrate above.

Proof of Main Theorem Pt. 4

We now have,

$$\theta(t_2) - \theta(t_1) = \int_{t_1}^{t_2} \int_{\mathcal{M}_t} \left| \vec{H} + \frac{(x - x_0)^{\perp}}{2(T - t)} \right|^2 \Phi(x, t) \mathrm{dVol}\ dt.$$

Changing variables to the rescaled setting for $a < b < 0$ yields

$$\theta_{x_0,T}(T + \lambda_k^{-2} b) - \theta_{x_0,T}(T + \lambda_k^{-2} a) =$$

$$\int_a^b \int_{\mathcal{M}_s^{(k)}} \left| \vec{H}^{(k)} - \frac{1}{2} \left\langle x, \nu^{(k)} \right\rangle \right|^2 \Phi(x) \mathrm{dVol}\ ds.$$

Observe when $\lambda_k \to \infty$, $t_1, t_2 \to T$. So, as $k_n \to \infty$,

$$\int_a^b \int_{\mathcal{M}_s^{(k)}} \left| \vec{H}^{(k)} - \frac{1}{2} \left\langle x, \nu^{(k)} \right\rangle \right|^2 \Phi(x) \mathrm{dVol}\ ds = 0,$$

for any $a < b < 0$. Hence, $\tilde{\mathcal{M}}_s$ satisfies $\tilde{\vec{H}} = \frac{1}{2} \langle x, \tilde{\nu} \rangle$ and it must be a self-shrinker.

Conclusion

Q&A